ENVS 423L/EPS 522 Problem Set #3

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, September 11 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of precipitation processes.

1.1 Humidity Calculations

Given the following air temperatures (T) and relative humidities (RH), calculate the vapor pressure (e) in kPa (5 pts) and the dew point temperature (T_d) in C (10 pts):

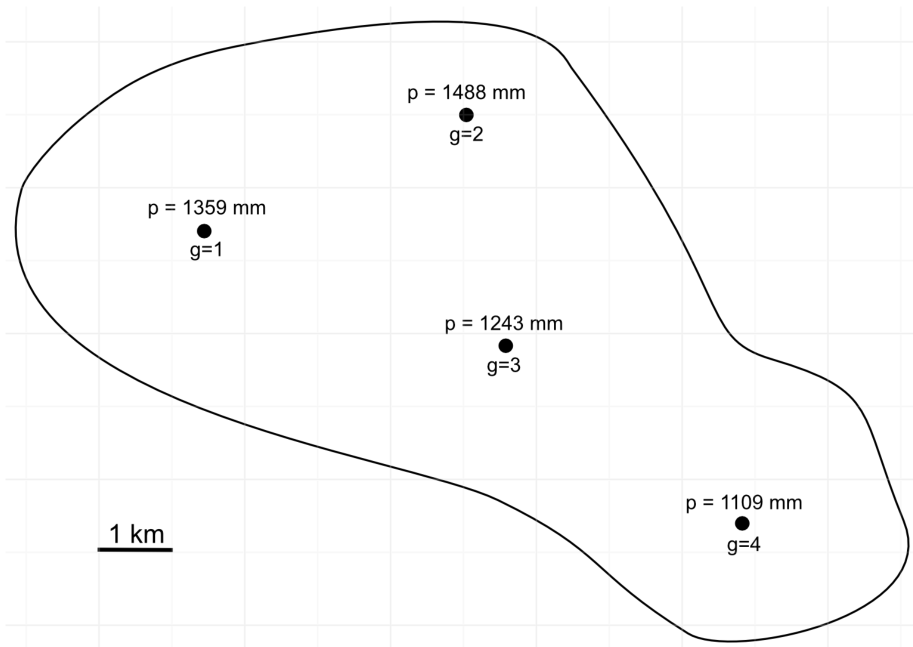
Quantity	A	B	C	D
T (˚C)	22	3	–7	29
RH (%)	53	74	100	22
e (kPa)				
T_d (˚C)				

1.2 Precipitable Water

Suppose 2 cm of rain fell from a nimbostratus cloud that is on average 700 m thick during the storm with an average water concentration of 0.5 g/m3. How much more rain fell (in terms of depth and ratio) than was in the cloud at that time? (10 pts)

1.3 Thiessen Polygons

Calculate the areal precipitation ($\hat{P}$) for the region below using the Thiessen polygon method. For partial grid cell intersection, just estimate down to the nearest 50%. (15 pts)



1.4 Hypsometric Method

Given the tables below, calculate the areal precipitation ($\hat{P}$) for the region using the hypsometric method. Hint: to predict $\hat{p}$ for an elevation range, assume the midpoint of the range as the value of z. (15 pts)

Elevation Range (m)	Fraction of Area
311–400	0.028
400–600	0.159
600–800	0.341
800–1000	0.271
1000–1200	0.151
1200–1400	0.042
1400–1600	0.008

Gage	Elevation (m)	Precip (mm)
1	442	1392
2	548	1246
3	736	1495
4	770	1698
5	852	1717
6	1031	1752

Compare this value to a simple arithmetic average and give a reason for the difference. (10 pts)

1.5 Depth-duration Frequency Analysis

Using the provided annual peak precipitation data for different durations for the Rivendell, M.E. station, perform a graphical frequency analysis using the Weibull plotting position. Plot (on the same plot) the log precipitation amount versus the log frequency (with frequency on a reversed x-axis) for each duration (15 pts)

What is the return period for a 10 mm 1-hr event and a 70 mm 24-hr event? (5 pts)

Discuss some limitations to this approach for water resources management. (10 pts)

1.6 Frequency Analysis and Stationarity

Cheng and AghaKouchak (2008) (pdf included) demonstrate the risk of not considering nonstationary trends in precipitation intensities by comparing stationary and nonstationary approaches for several gages across the US. From this paper, what main trend do they identify related to the underprediction of extreme rainfall using a stationary method? (7.5 pts)

If the 100-yr 1-hr and 2-hr intensity for the White Sands station predicted using stationary methods is 40 and 25 mm/hr, respectively (shown in Figure 1), what is the underestimation (as predicted by their nonstationary approach) in terms of percent difference? Hint: percent difference is difference divided by original value times 100. The differences for the 100-year 1- and 2-hr durations should be taken as the median value of the boxplots shown in Figure 2. (7.5 pts)